

Open Text Metrics Overview

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Abstract

We investigate the impact of data quality filtering on various open text metrics within a large dataset (1.9 million) of context-question-answer generated from Arizona State University (ASU) website data. Given the subjective nature of "good" and "bad" answers and the impracticality of manual scoring, we analyze how applying a threshold to the bottom 50% of data, as defined by different metrics (similarity score, bleu, Cosine similarity, BERT similarity, character F-score, rouge 1, rouge 2, rougeL), impacts the minimum and median values of metrics. The analysis reveals that different metrics exhibit varying sensitivities to these cutoffs. F-Score Similarity demonstrates the greatest impact on minimum values, while Rouge 2 shows the largest impact on median values. NLTK Bleu and Sacre Bleu display erratic behavior, potentially due to their deeply skewed distributions. The findings inform the selection of a robust "North Star" metric for future data filtering, balancing the need to remove low-quality data with the preservation of overall data integrity.

Goal

Selection of a metric for filtering lower-quality data, as defined by various open text metrics, and the resulting impact on the remaining distributions. The goal is to inform the selection of a robust metric for future data filtering and a generation of a ground truth dataset that can be utilized in training models.

Methods

Data

There are 1,996,139 usable records from the ASU website scrape, these were generated in the following way:

- Each webpage was broken into 500-word increments (contexts) with a 20 word overlap.
- Then each 500-word increment had 3 questions generated from AI
- Each question was responded to in 3 different ways (due to the stochastic nature of the responses from the generative AI)

Therefore each 500 word increment of a webpage has 9 context-question-answer pairs.

Figure 1: Distribution of the original ASU webpage word count, which is heavily skewed.

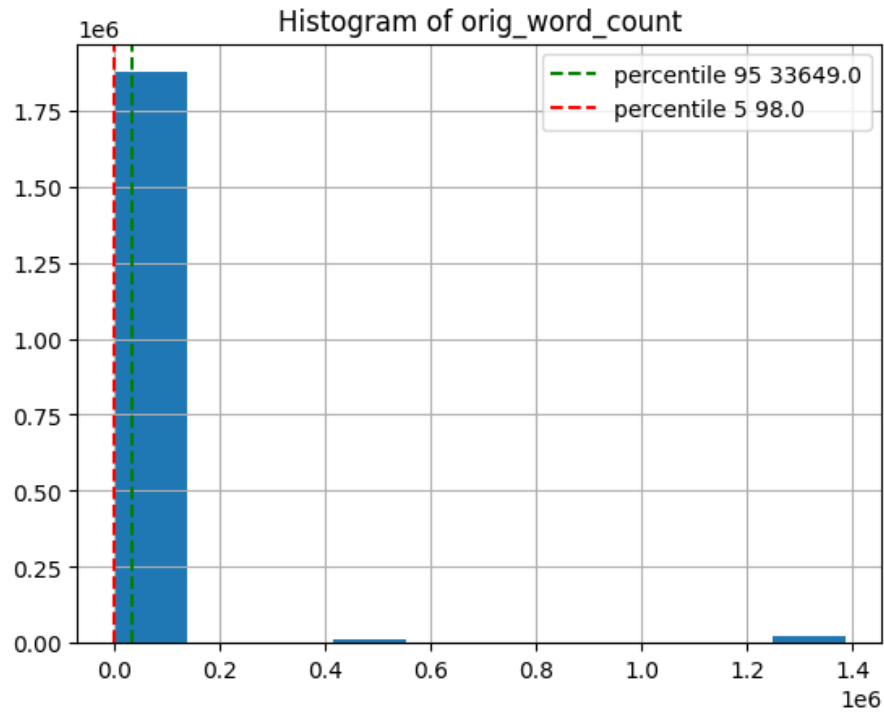
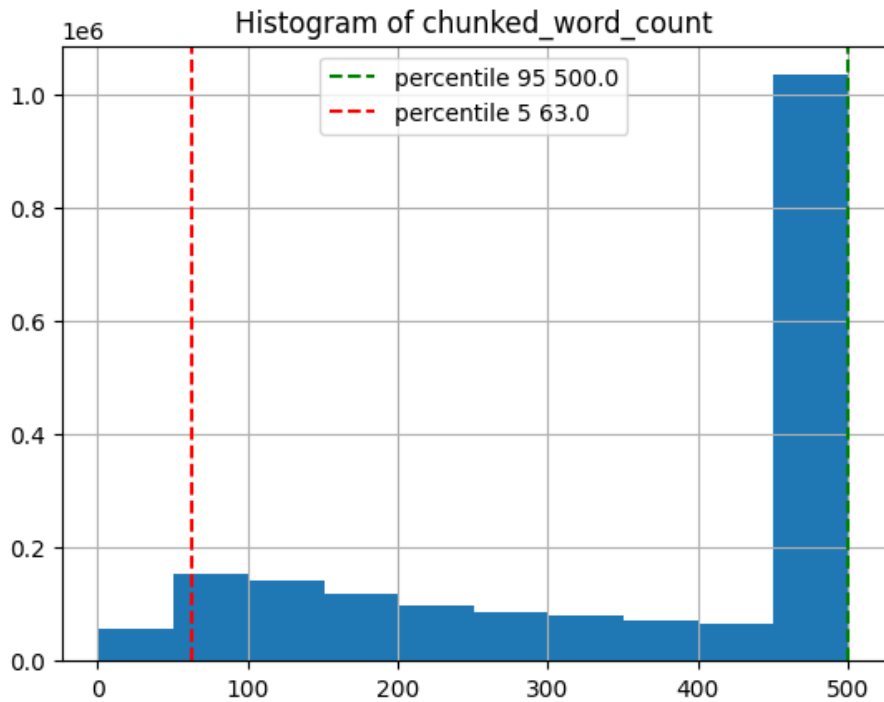


Figure 2: Chunked word count for ASU webpages, in a skewed distribution.

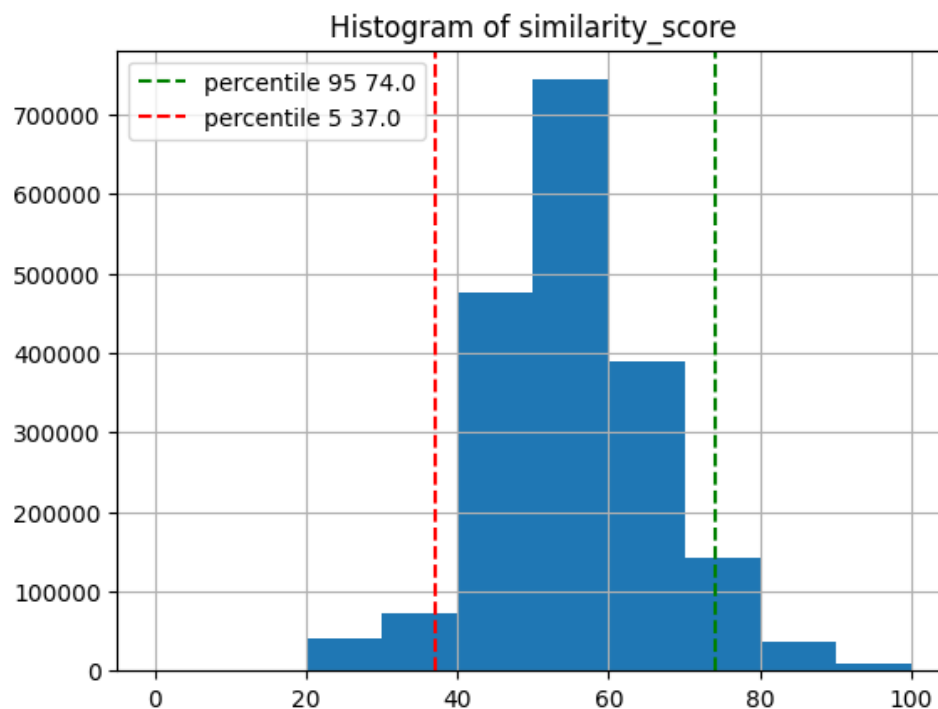


Each question and answer generated are not equally “good.” While defining what is a “good” answer vs a “bad” answer is very subjective and therefore changes based upon the individual. As well as I do not believe any individual wants to read 1.9 million questions, answer pairs and score them. As such looking at the impact of different cutoffs around several metrics (to pick the “best” metric) so we feel more confident in defining a North Star metric to move forward on in the future. The metrics that were considered are as follows (similarity score, NLTK bleu, sacrebleu bleu, Cosine, sBURT, FSCORE SIM, rouge1, rouge2, rougeL, rougeLsum) and their definitions are provided in the next section.

Definitions

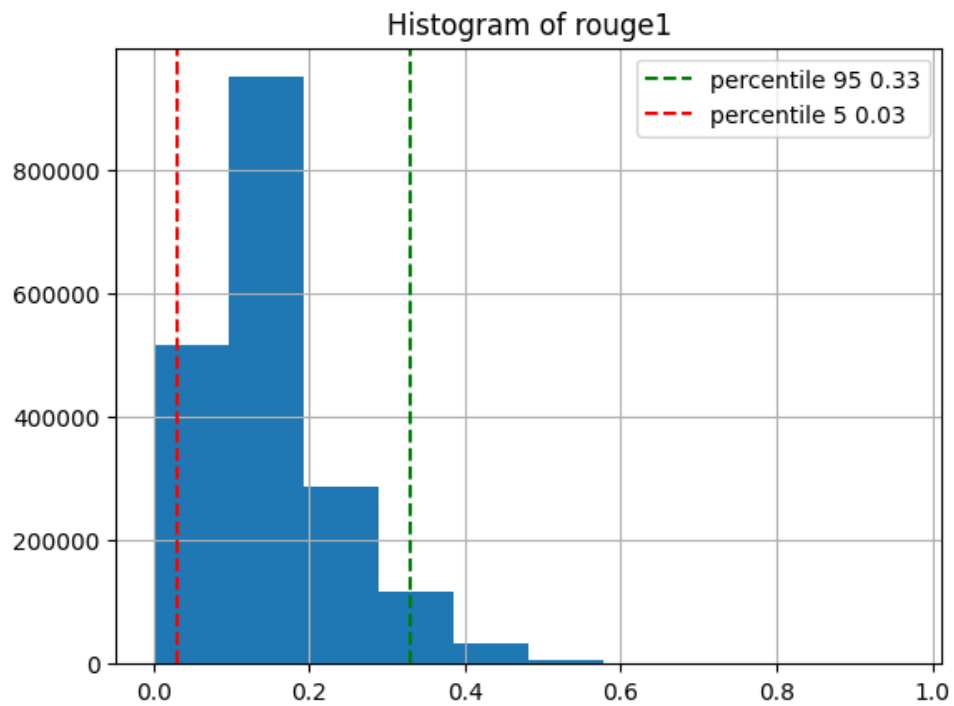
Similarity Score (fuzzywuzzy): This metric is calculated using the fuzzywuzzy library, specifically the fuzz.partial_ratio function. partial_ratio finds the largest matching subsequence and computes the ratio of its length to the length of the shorter string. This makes it useful for cases where the strings might have different lengths or contain extra information. The score ranges from 0 to 100, with higher scores indicating a greater degree of fuzzy similarity between the response and the context.

Figure 3: Distribution of similarity score (fuzzywuzzy) on 1.9 million records of context-question-answer, appears to be normally distributed.



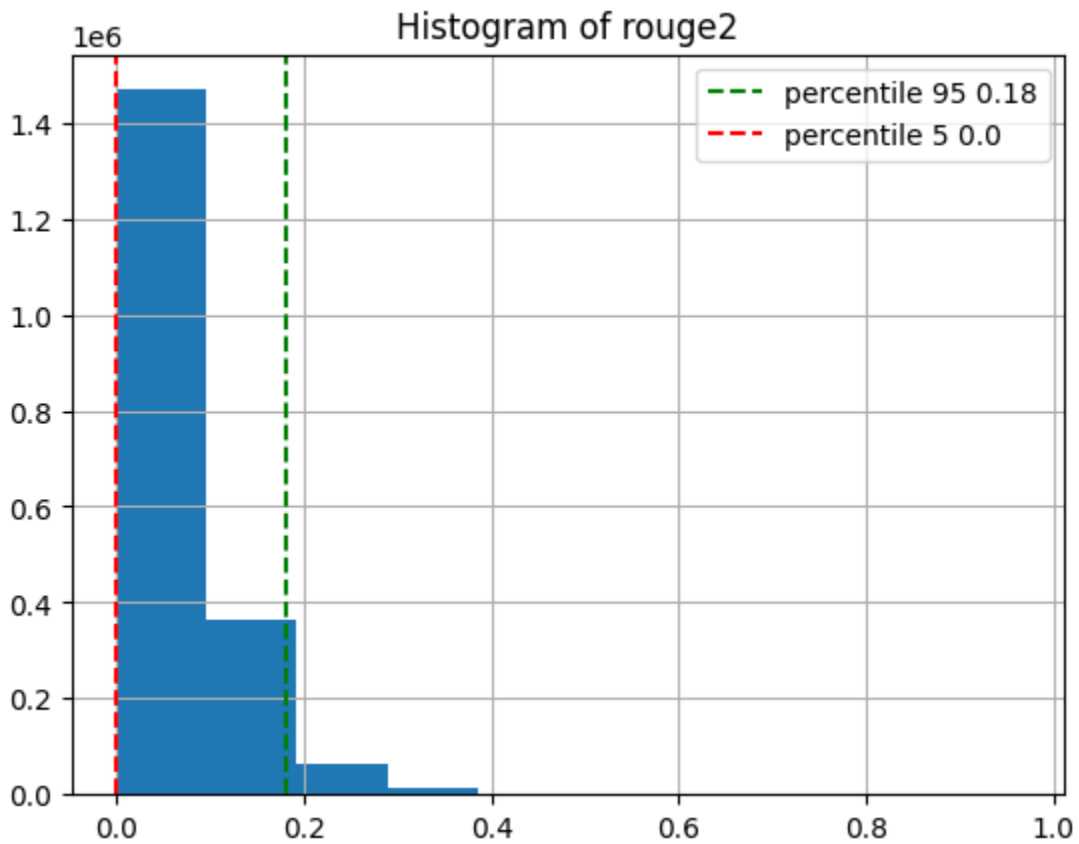
ROUGE-1: Measures how many individual words (unigrams) are shared between a generated summary and a reference summary. A higher score means more words are shared.

Figure 4: : Distribution of rouge 1 on 1.9 million records of context-question-answer



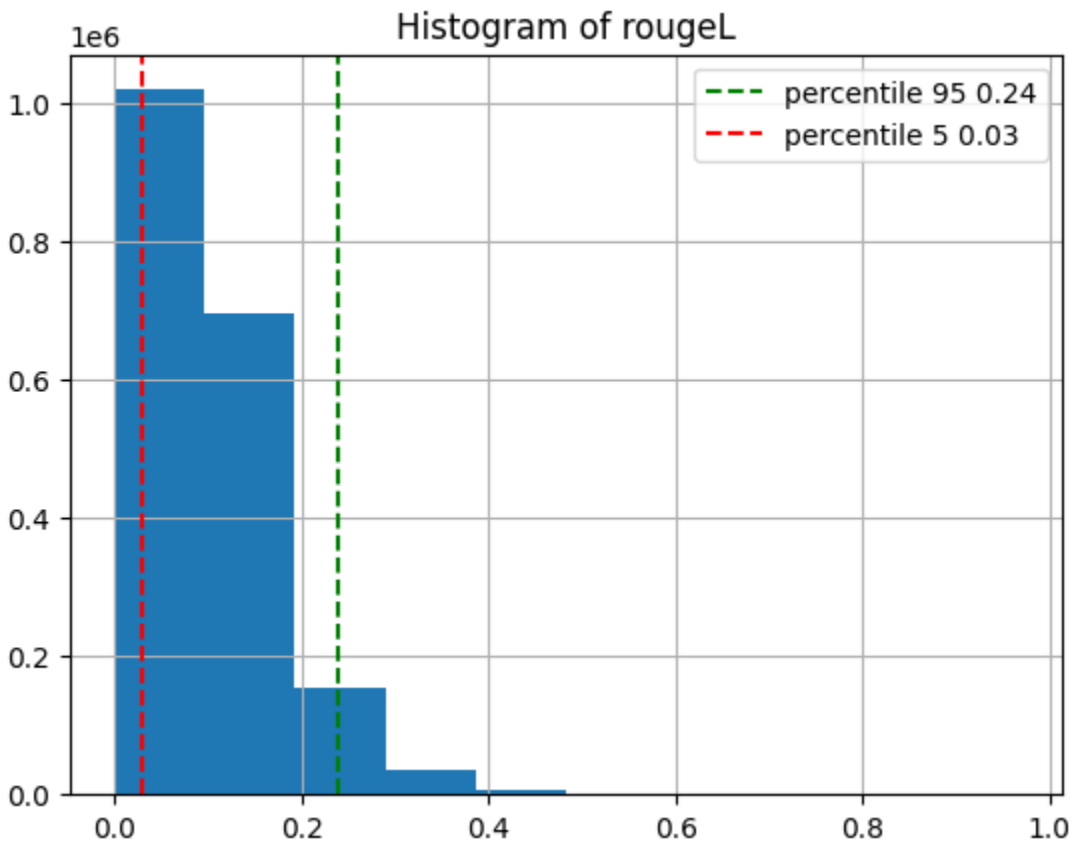
ROUGE-2: Measures how many pairs of consecutive words (bigrams) are shared between a generated summary and a reference summary. A higher score means more consecutive word pairs are shared.

Figure 5: : Distribution of rouge 2 on 1.9 million records of context-question-answer



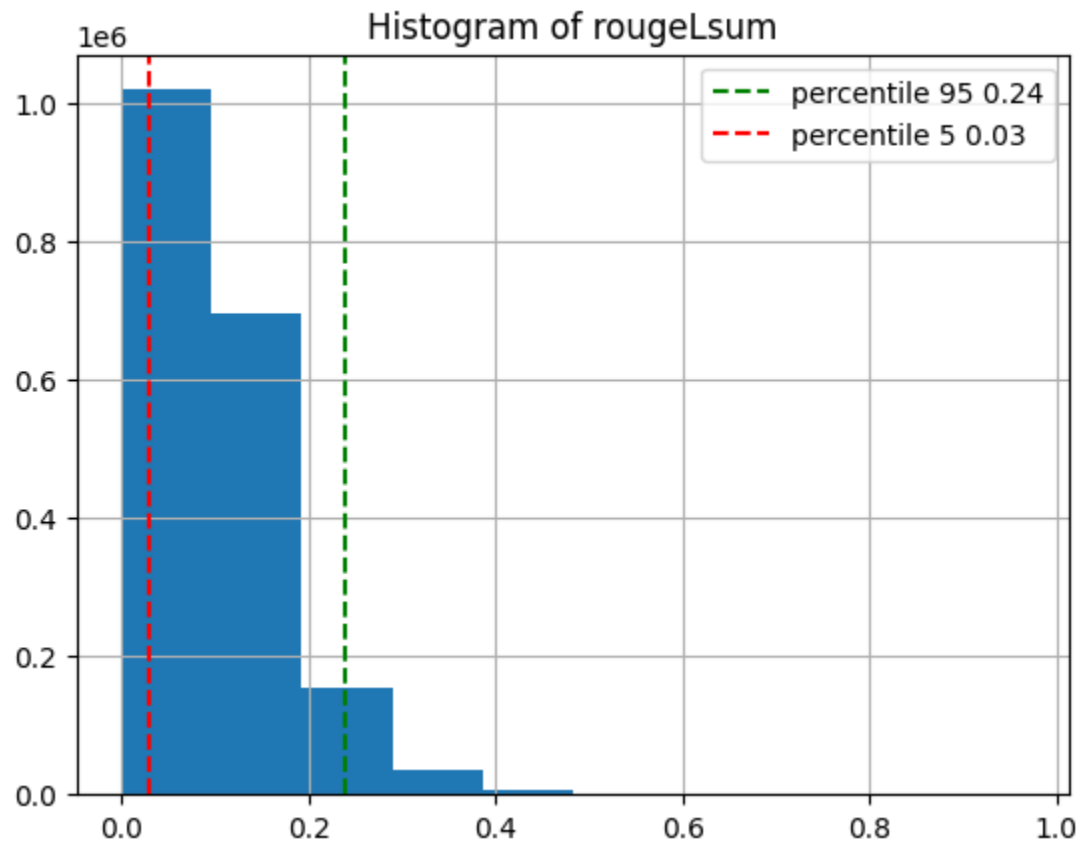
ROUGE-L: Measures the longest sequence of words that appear in both the generated summary and the reference summary. A higher score means the longest common sequence is longer.

Figure 6: : Distribution of rouge-L on 1.9 million records of context-question-answer



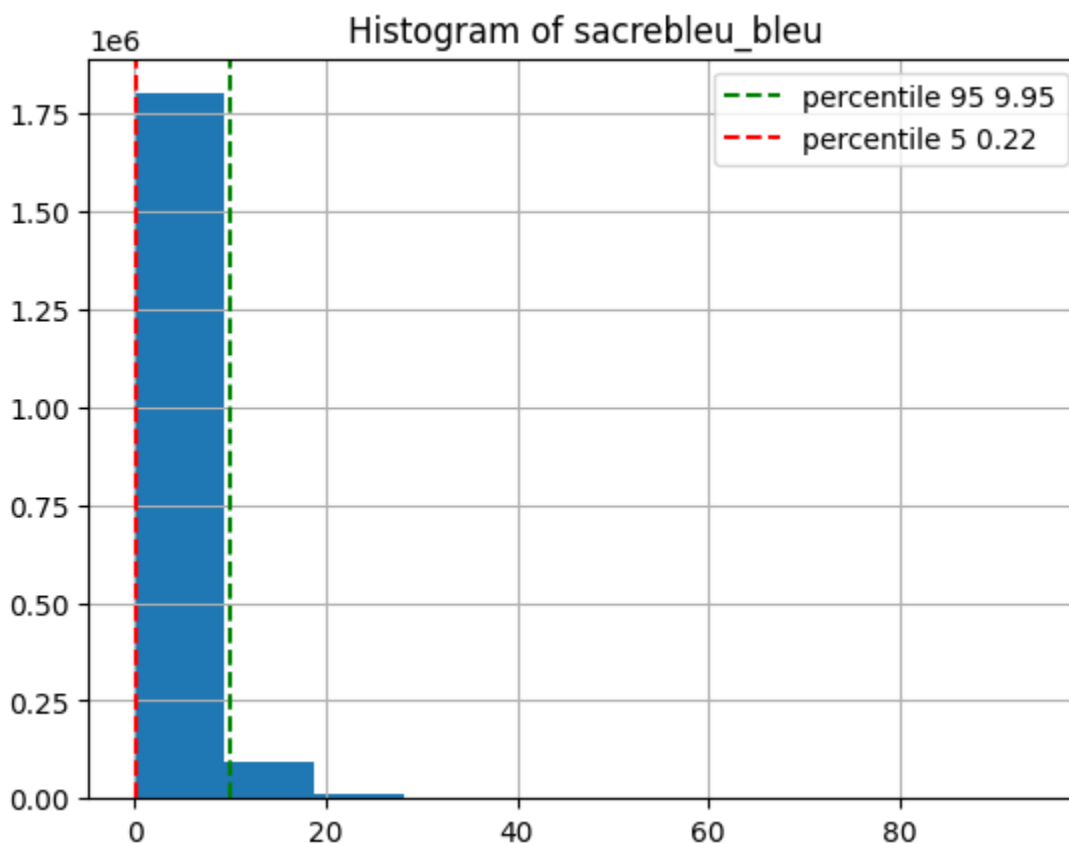
ROUGE-Lsum: Similar to ROUGE-L, but it calculates the longest common word sequences for each sentence and then averages the results. This is useful for tasks where sentence-level similarity is important. A higher score means more sentence-level similarity.

Figure 7: : Distribution of rouge-Lsum on 1.9 million records of context-question-answer



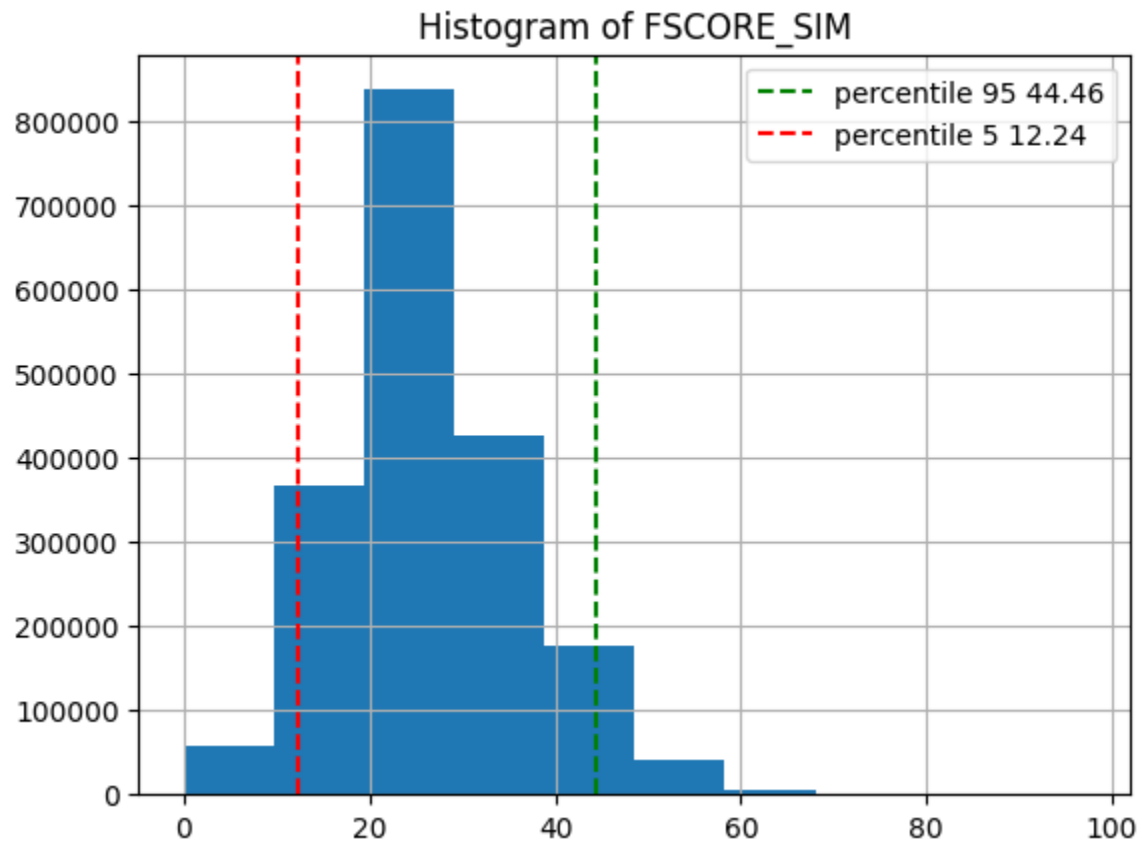
Bilingual Evaluation Understudy (BLEU)(sacrebleu): Measures the similarity of a machine-translated text to one or more reference translations by counting matching word sequences from the sacrebleu library. A higher score indicates better translation quality.

Figure 8: : Distribution of BLEU (sacrebleu) on 1.9 million records of context-question-answer



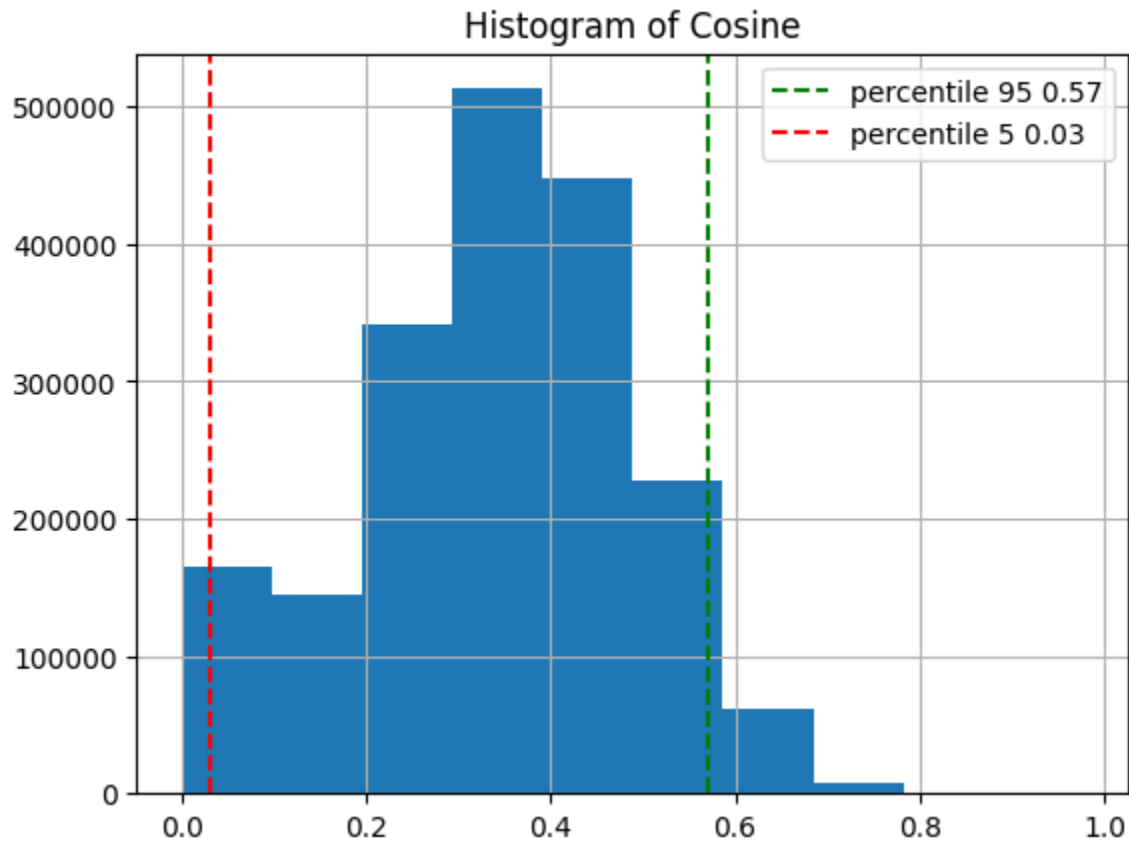
CHRF (Character n-gram F-score): Measures the similarity between a generated text and a reference text by looking at the overlap of character sequences. A higher score indicates greater similarity.

Figure 9: : Distribution of CHRF on 1.9 million records of context-question-answer



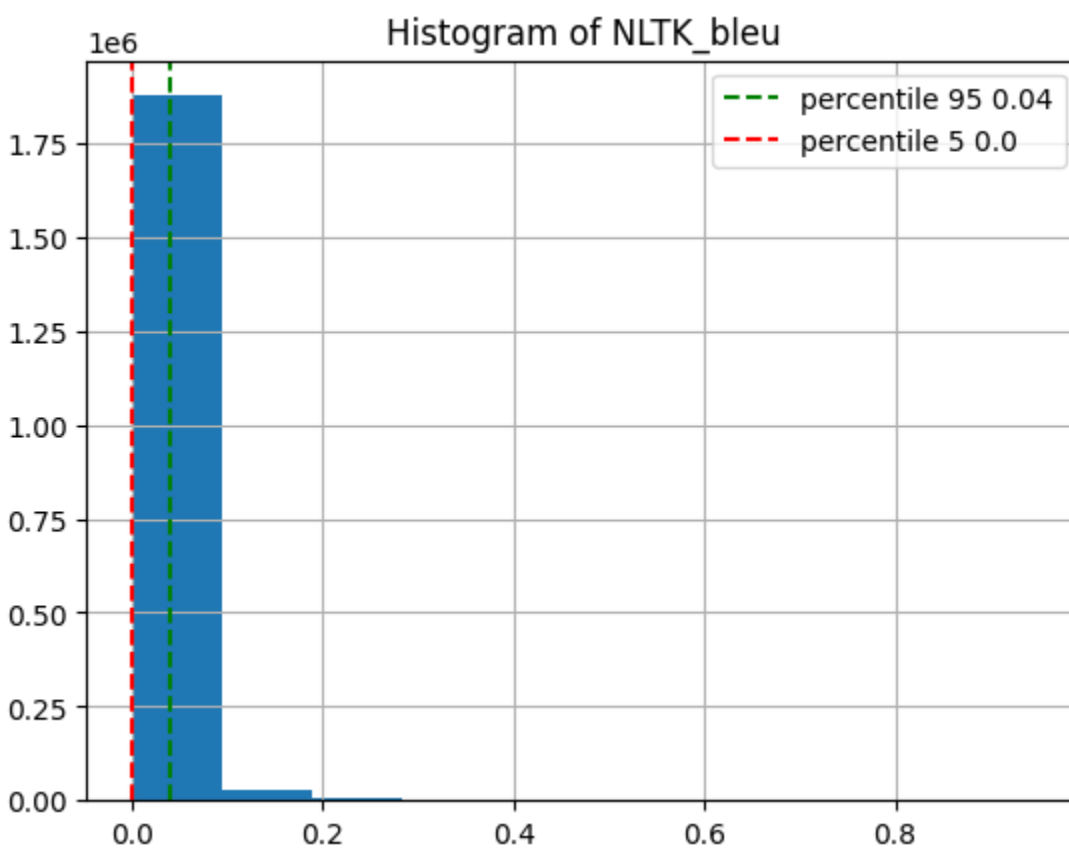
TF-IDF Cosine Similarity: Measures the similarity between two texts by comparing the cosine of their TF-IDF vector representations. TF-IDF weighs words by their frequency in the document and across all documents. A higher score (closer to 1) indicates greater similarity.

Figure 10: : Distribution of cosine similarity on 1.9 million records of context-question-answer



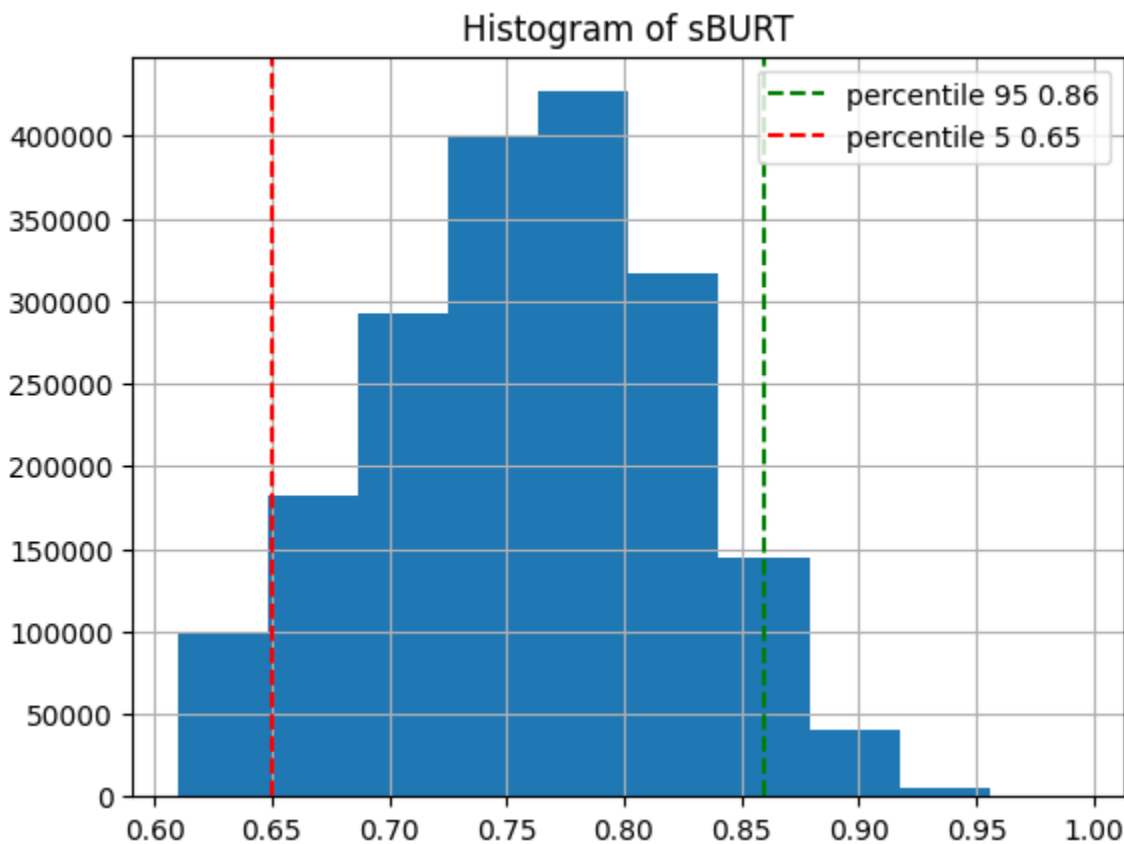
NLTK BLEU: The BLEU score calculation as implemented in the NLTK library. Measures the similarity of a generated text to one or more reference translations. A higher score indicates better translation quality.

Figure 11: : Distribution of NLTK bleu on 1.9 million records of context-question-answer



sBERT Similarity (sBERT): Measures the semantic similarity between two texts using Sentence-BERT (sBERT) embeddings. BERT (Bidirectional Encoder Representations from Transformers) is a transformer-based neural network model that learns to represent the meaning of words in context. sBERT is a modification of BERT that is optimized for generating sentence embeddings. The embeddings capture the meaning of the input texts, and cosine similarity measures the similarity between these embeddings. A higher score (closer to 1) indicates greater semantic similarity.

Figure 12: : Distribution of sBERT on 1.9 million records of context-question-answer



Additional metrics from the Ragas package were considered, however these are very time-consuming to calculate and not feasible to implement on 1.9 million records.

Data is available here:

https://huggingface.co/datasets/onyxsarah/ASU_website/blob/main/6_scored.csv

Approach

The following steps were applied:

1. Apply a 50% threshold to a metric
2. Look at the impact in the other metrics

For example, a 50% threshold was applied to Fuzzywuzzy Similarity Score, so the cut off was 54. This becomes our filtered data, and we look at the resulting impact on the other metrics (NLTK bleu, sacrebleu bleu, Cosine, sBURT, FSCORE SIM, rouge1, rouge2, rougeL, rougeLsum) for their minimum value and median value.

The results for the minimum and median value for each 50% threshold applied is reported in table 1 and 2. The values reported are the percentile change in the minimum and median. The following equation was used to calculate the percent change:

$$percentile = \frac{(threshold - initial\ threshold)}{initial\ threshold} * 100$$

Where:

- threshold is the minimum/median value from the filtered data
- initial threshold is the minimum/median value from the unaltered data

Results

Table 1: The percent change on the minimum value given a 50% threshold was applied to the metric.

metrics	similarity_score	NLTK_bleu	sacrebleu_bleu	Cosine	sBURT	FSCORE_SIM	rouge1	rouge2	rougeL	rougeLsum
similarity_score	53.855	0.000	0.000	0.000	0.000	0.084	0.003	0.003	0.003	0.003
NLTK_bleu	0.004	50.000	0.000	0.000	0.000	0.022	0.003	0.003	0.003	0.003
sacrebleu_bleu	0.150	0.000	50.112	0.000	0.000	0.199	4.281	0.003	5.109	5.109
Cosine	0.059	0.000	0.000	50.000	0.000	0.067	4.423	0.003	4.670	4.670
sBURT	0.000	0.000	0.000	0.000	50.000	0.000	0.003	0.003	0.003	0.003
FSCORE_SIM	0.566	0.000	0.000	0.000	0.000	50.014	0.003	0.003	0.003	0.003
rouge1	0.054	0.000	0.000	0.000	0.000	0.067	50.004	0.003	6.564	6.564
rouge2	0.069	0.000	0.000	0.000	0.000	0.067	10.944	50.020	6.914	6.914
rougeL	0.054	0.000	0.000	0.000	0.000	0.067	25.242	0.003	50.023	50.023
rougeLsum	0.054	0.000	0.000	0.000	0.000	0.067	25.242	0.003	50.023	50.023

Table 2: The percent change on the median value given a 50% threshold was applied to the metric.

metrics	similarity_score	NLTK_bleu	sacrebleu_bleu	Cosine	sBURT	FSCORE_SIM	rouge1	rouge2	rougeL	rougeLsum
similarity_score	14.815	26.054	53.540	17.790	2.064	8.511	10.708	41.929	18.289	18.289
NLTK_bleu	0.000	6231.765	68.584	6.268	3.543	28.318	45.933	53.475	45.049	45.049
sacrebleu_bleu	9.259	1454.856	83.186	16.936	3.597	23.397	34.472	61.493	39.412	39.412
Cosine	7.407	80.326	42.920	27.063	2.436	9.964	14.243	34.829	17.058	17.058
sBURT	3.704	904.227	44.690	11.210	5.928	17.789	27.571	35.371	27.309	27.309
FSCORE_SIM	3.704	3861.377	76.549	13.363	3.991	28.439	43.920	58.784	43.481	43.481
rouge1	3.704	4646.751	78.761	14.568	4.183	28.318	45.933	61.493	45.846	45.846
rouge2	9.259	1581.390	82.301	18.123	3.701	24.002	36.447	63.050	41.020	41.020
rougeL	7.407	4455.579	81.416	15.199	4.095	27.914	45.678	62.462	45.846	45.846
rougeLsum	7.407	4455.579	81.416	15.199	4.095	27.914	45.678	62.462	45.846	45.846

Conclusions

In summary, NLTK Bleu and Sacre Bleu exhibit steeply skewed distributions, with NLTK Bleu demonstrating more discrete behavior and consequently, erratic sensitivity to data filtering. RougeL and RougeLsum are strongly correlated, suggesting redundancy. F-Score Similarity has the greatest influence on the minimum values of other metrics, while Rouge 2 most significantly impacts their median values. Therefore, to create a high-quality "golden" dataset, data filtering should prioritize Rouge 2 and F-Score Similarity.